

SpyGlass Lint & CDC Waiver Report

UART-Controlled Low-Power Digital System

Scope	Lint, Clock/Reset Integrity, CDC Verification and CDC Structural Review
Tool	Synopsys SpyGlass
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This report documents reviewed SpyGlass violations that were intentionally waived after design-level analysis. Each waiver is tied to a specific implementation choice such as clock gating, clock division, asynchronous FIFO synchronization, or stable UART configuration control.

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1. Project and Verification Scope

The design is a UART-controlled digital system operating across multiple clock domains. The REF_CLK domain contains the register file, ALU, system controller and clock-gating logic, while UART processing uses derived RX/TX clocks. CDC protection is implemented using synchronizers and an asynchronous FIFO.

The purpose of this report is not to hide violations. It records cases where SpyGlass correctly raised a rule, the RTL structure was reviewed, and the remaining violation was judged intentional or structurally protected by the architecture.

Review flow

1) Run Lint and CDC goals. 2) Inspect each rule and source location. 3) Decide whether the issue requires an RTL fix or is intentional. 4) Apply only justified waivers. 5) Re-run and archive waiver statistics.

Verification Areas

Area	Primary purpose
Lint	RTL coding-style and synthesis-quality checks
Clock / Reset Integrity	Clock glitch and clock-control structure checks
CDC Verify	Functional CDC safety and data-stability analysis
CDC Verify Struct	Structural CDC topology analysis

2. Waiver Statistics Summary

The following values are taken directly from the generated SpyGlass waiver reports.

Goal / Report	Generated	Waived	Reported	Applied Waivers
CDC Verify	58	5	53	4
CDC Verify Struct	50	1	49	1
Clock / Reset Integrity	11	2	9	1

Lint Waiver Set

Rule	Waived messages / scope	Reason
InferLatch	CLK_GATE	Intentional level-sensitive latch for glitch-free clock gating
STARC05-1.4.3.4	RX and TX clock dividers	Generated clock toggles using its own registered state
STARC05-2.11.3.1	edge/bit counter logic	Counter control was classified by the rule as FSM-like logic

Interpretation

A waiver is accepted only where the RTL intent is understood and the implementation contains a separate mechanism that guarantees safe behavior. Waivers are not used as a substitute for fixing genuine design defects.

3. Lint Waivers

3.1 Intentional Clock-Gating Latch

Rule	InferLatch
Severity	Error
Tool finding	SpyGlass reports an inferred latch for Latch_Out inside CLK_GATE.
Technical review	The latch is deliberate. It stores CLK_EN while the source clock is low so the enable remains stable while the gated clock is high, which is the intended glitch-free clock-gating structure.
Waiver comment	Intentional latch used for Clk_Gate

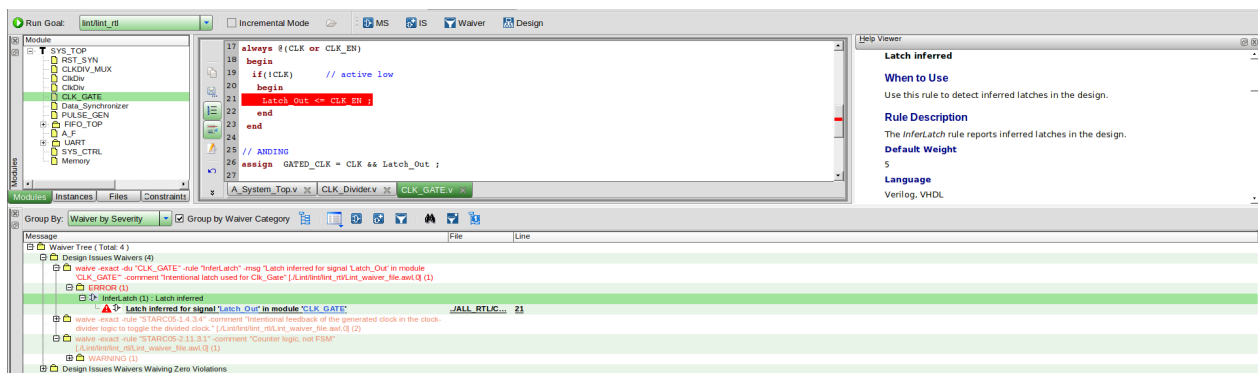


Figure 1: *InferLatch* waiver in *CLK_GATE*.

3.2 Generated Clock Toggle Logic

Rule	STARCO5-1.4.3.4
Severity	Warning
Tool finding	The clock-divider output <code>div_clk</code> is also read as a data value in the toggle assignment <code>div_clk <= !div_clk</code> .
Technical review	The feedback is local to the clock-divider register and is required to toggle the generated clock. The same pattern appears in both RX and TX clock dividers.
Waiver comment	Intentional clock-divider toggle

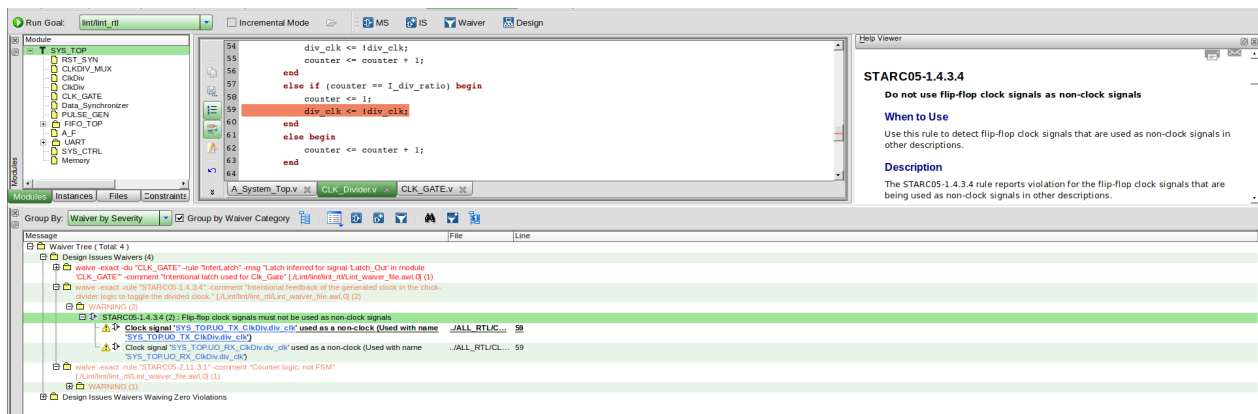


Figure 2: STARCO5-1.4.3.4 warning for RX/TX clock-divider toggle logic.

3.3 Counter Logic Misclassified as FSM

Rule	STARCO5-2.11.3.1
Severity	Warning
Tool finding	The rule reports sequential and combinational parts of an FSM in the same always block around the bit/edge counter logic.
Technical review	The reviewed block is counter/control logic, not the UART state-machine implementation. The rule pattern is triggered by the counter update and conditional control structure.
Waiver comment	Counter logic, not FSM

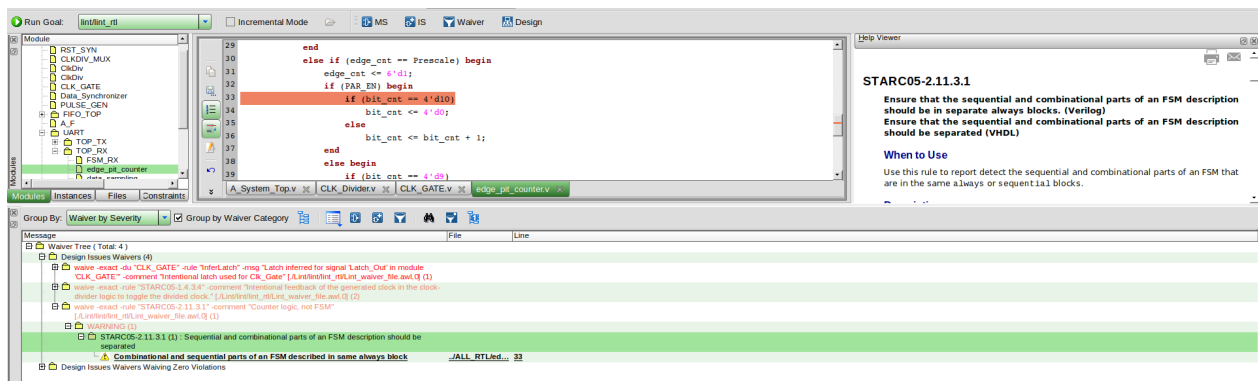


Figure 3: STARCO5-2.11.3.1 warning on edge/bit counter logic.

4. Clock / Reset Integrity Waiver

Rule	Clock_glitch04
Severity	Warning (2 messages)
Tool finding	SpyGlass finds a generated clock and a configuration-register bit converging through combinational logic on a clock pin. One case is RX clock control through Prescale; the other is TX clock control through Div_Ratio.
Technical review	Prescale and Div_Ratio are configuration values and are kept static while the related UART clock is actively used. Under that operating assumption, the control path is not expected to change asynchronously during active UART operation.
Waiver comment	Static UART configuration during operation

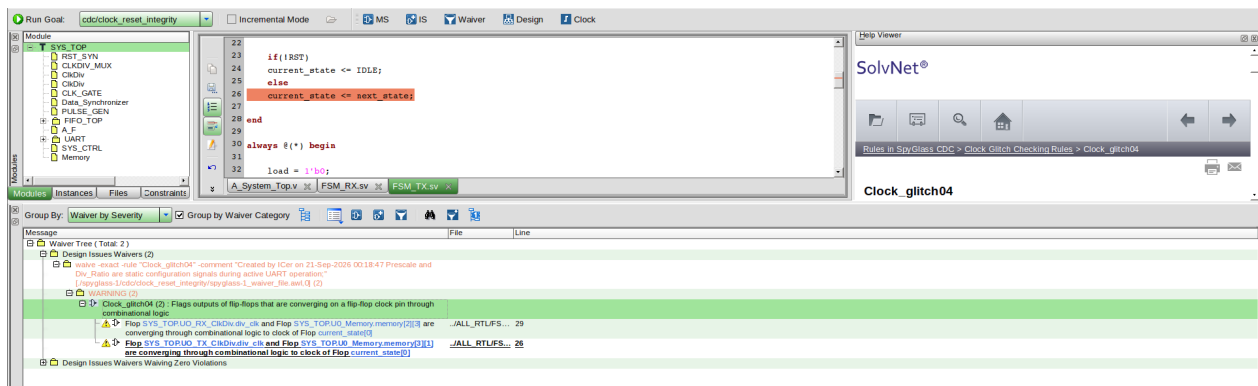


Figure 4: Clock_glitch04 waiver tree showing two waived clock-control messages.

5. CDC Verification Waivers

5.1 FIFO Data-Bus Synchronization Scheme

Rule	Ac_conv04
Severity	Warning
Tool finding	SpyGlass reports that bits of the FIFO memory source bus are not uniformly synchronized to the UART TX destination bus because a different synchronization scheme is used.
Technical review	An asynchronous FIFO does not synchronize every payload bit with individual 2-FF synchronizers. Data validity is controlled by synchronized Gray-coded pointers together with FIFO empty/read-control logic, so the payload is sampled only when valid in the read domain.
Waiver comment	FIFO protected by Gray-pointer control

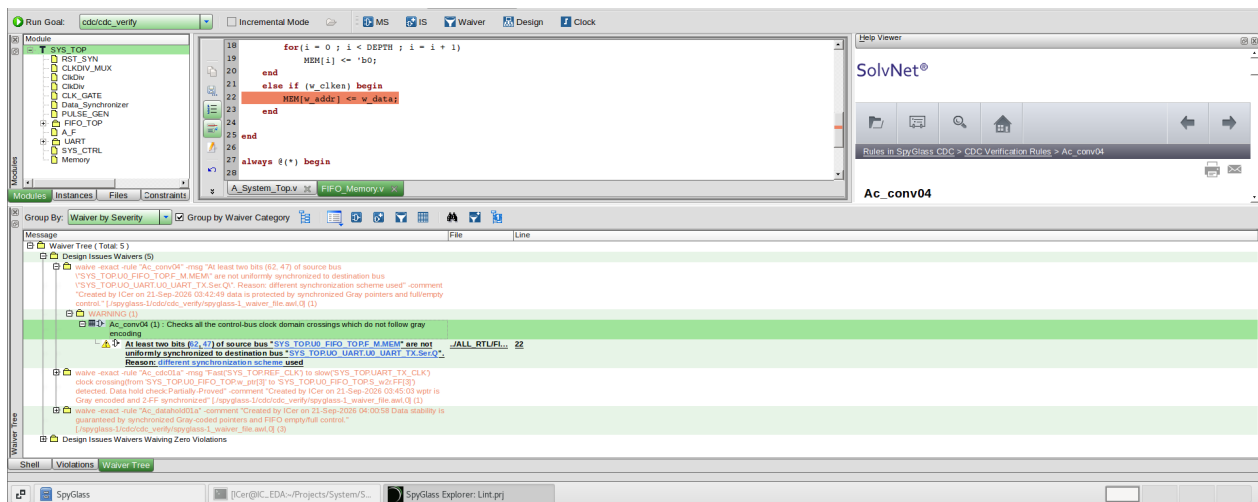


Figure 5: *Ac_conv04* warning on the asynchronous FIFO payload crossing.

5.2 Gray Pointer Fast-to-Slow Crossing

Rule	Ac_cdc01a
Severity	Warning
Tool finding	SpyGlass reports a fast REF_CLK to slower UART_TX_CLK crossing on the write pointer with the data-hold check marked Partially-Proved.
Technical review	The write pointer is Gray encoded before crossing and then passes through a two-flop synchronizer in the destination domain. Gray coding limits each pointer transition to one bit, which is the intended async-FIFO synchronization method.
Waiver comment	Gray pointer + 2-FF synchronization

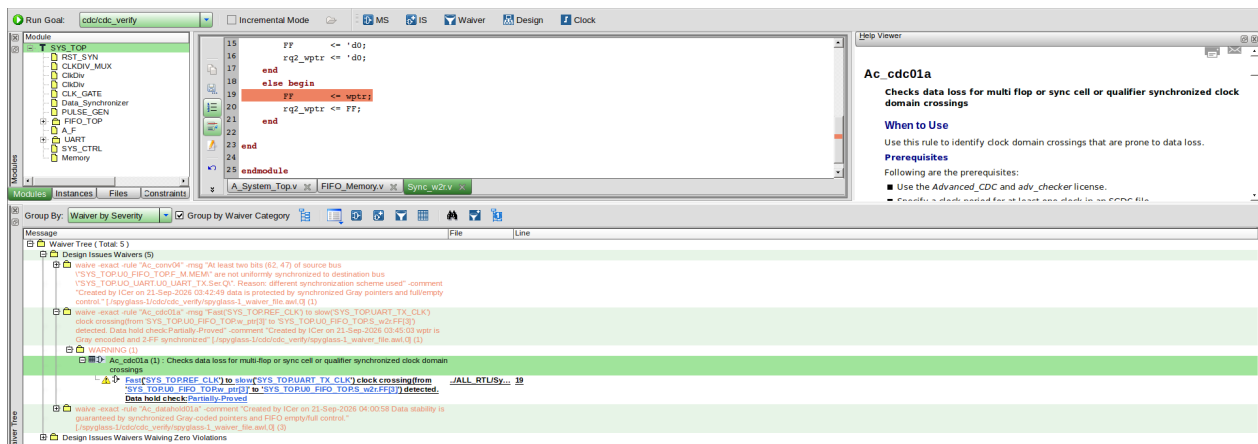


Figure 6: *Ac_cdc01a* fast-to-slow Gray-pointer crossing.

5.3 Data Stability / Hold Checks

Rule	Ac_datahold01a
Severity	2 Warnings + 1 Info
Tool finding	SpyGlass checks synchronized data crossings from FIFO memory to UART TX logic and from UART RX data to the REF_CLK domain. Two data-enable sequencing checks are Partially-Proved and one check is PASSED.
Technical review	FIFO payload stability is tied to synchronized Gray-pointer control and empty/full sequencing. The RX-side Data_Synchronizer path is explicitly reported as PASSED by the tool.
Waiver comment	Data stable under FIFO control

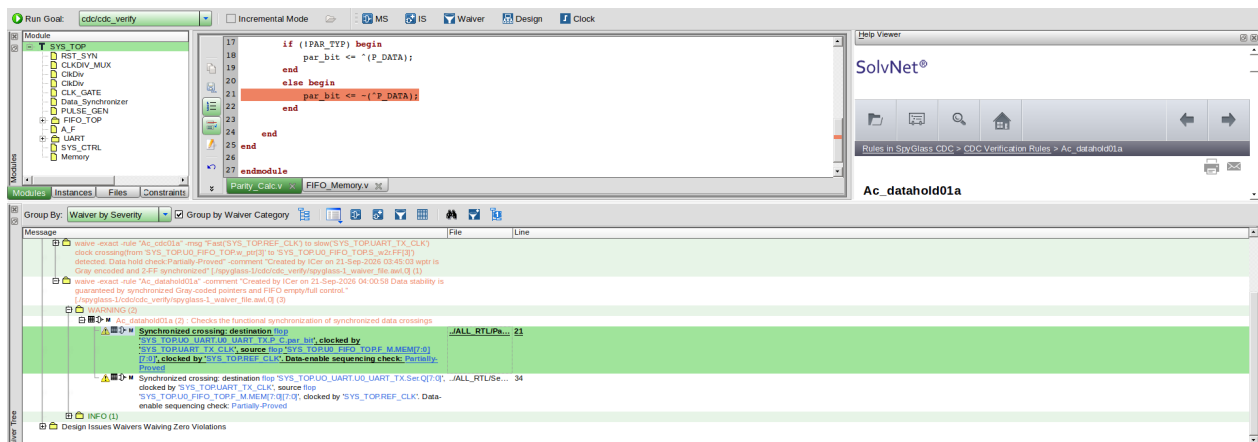


Figure 7: *Ac_datahold01a* data-stability checks in the CDC waiver tree.

5.4 Structural CDC: FIFO Empty Comparison

Rule	Ac_conv02
Severity	Warning
Tool finding	SpyGlass reports that synchronized write-pointer bits converge on <code>FIFO_EMPTY</code> while the Gray-encoding check is shown as disabled.
Technical review	The write pointer is Gray encoded and is synchronized through a 2-FF synchronizer before it participates in the FIFO empty comparison. The warning is therefore reviewed against the intended asynchronous-FIFO architecture.
Waiver comment	Gray pointer synchronized before empty comparison

Report evidence
CDC Verify Struct: 50 generated messages, 1 waived message, 49 reported messages, and 1 applied waiver.

Why this waiver is separate

This check is structural rather than a payload data-hold check. It focuses on multiple synchronized pointer bits converging into `FIFO_EMPTY`. The waiver documents the design intent that the converging bus is a Gray-coded pointer that has already been synchronized.

6. Final Assessment

The reviewed waivers fall into four implementation categories: intentional clock-gating structure, intentional generated-clock toggle logic, a lint rule pattern on counter logic, and asynchronous-FIFO / UART CDC structures.

No waiver in this report is justified only by the fact that the warning is inconvenient. Each one is tied to an explicit architectural condition: a latch intentionally used for clock gating, a registered clock divider, static UART configuration values, or Gray-coded pointer synchronization and FIFO control.

Final status

All documented violations were reviewed and intentionally waived with a technical reason. The waiver files and SpyGlass waiver reports should be kept with the verification deliverables so the reasoning remains traceable.

REVIEWED / JUSTIFIED WAIVERS

Documented Waiver Set

Rule	Area	Disposition
InferLatch	Lint	Intentional clock-gating latch
STARC05-1.4.3.4	Lint	Intentional clock-divider toggle
STARC05-2.11.3.1	Lint	Counter logic, not FSM
Clock_glitch04	Clock/Reset Integrity	Static Prescale / Div_Ratio during active UART
Ac_conv04	CDC Verify	Async FIFO payload protected by pointer/control scheme
Ac_cdc01a	CDC Verify	Gray pointer + 2-FF synchronization
Ac_datahold01a	CDC Verify	Data stability controlled by FIFO / synchronized control
Ac_conv02	CDC Verify Struct	Gray pointer synchronized before FIFO empty comparison